

SUMMER WORKSHOP IN MATHEMATICS

The Probabilistic Method

A five-day workshop for high-school students



Workshop Notes
(Lectures, Worked Examples, and Problems)

City of Lights
May 25, 2026

“The probabilistic method has now developed to a point where it would be hopeless even to attempt to give a complete coverage of the area in one paper.” — Paul Erdős

About these notes. This document accompanies a five-day workshop on the probabilistic method aimed at mathematically curious 9th–10th standard students. Each chapter is one full day’s worth of material (about three hours of class time) and ends with a problem set that students take home, work on overnight, and return the next morning.

The problem sets are also available as two companion documents: `WorkshopProblems.pdf` (problems only, for day-by-day distribution to students) and `WorkshopSolutions.pdf` (the same problem sets, with worked solutions).

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CHAPTER 1

DAY 1 — FOUNDATIONS OF PROBABILITY

The goal of this first day is to build, from scratch, the language we will need for the rest of the week: *sample spaces*, *events*, and *probability measures*. Although the workshop is about a powerful proof technique — the probabilistic method — the technique relies on a small handful of ideas that we now make precise.

A piece of advice before we begin. The single most common error in elementary probability is to start computing before deciding *what experiment* one is computing about. We will be obsessive about this. Every probability we compute will be tied to an explicit sample space and an explicit probability measure — this discipline is what allows us, later in the week, to use probability to prove results that have nothing to do with chance at all.

1.1 Why we need a formal language. Before we write down any definitions, let us see what goes wrong when we do not. Suppose two students are asked: *a fair coin is tossed twice; what is the chance that both tosses show heads?* The first student reasons, “There are three possible outcomes — two heads, two tails, or one of each — so the answer is $1/3$.” The second student insists, “No, there are four equally likely outcomes — HH, HT, TH, TT — so the answer is $1/4$.” Both arguments sound plausible if read quickly. Both cannot be correct. Yet neither student has done anything obviously foolish: each has listed what they take to be “the possible outcomes” and counted favourable ones. The disagreement is not about arithmetic. It is about *what experiment is being performed in the first place*, and which list of outcomes is the honest one. Once we agree to record the two tosses in order (so the four outcomes HH, HT, TH, TT form the right list, and each is equally likely), the answer is forced: $1/4$. The first student’s list was not wrong because the arithmetic was wrong; it was wrong because it grouped two genuinely different outcomes — the one where the first toss was a head and the second a tail, and the reverse — into a single entry, while still pretending each entry was equally likely.

This small disagreement is the whole story in miniature. The phrase “the probability of X ” is, on its own, only half a sentence; it is completed only when we also commit to *which experiment* X refers to and *how likelihoods are assigned* to the outcomes of that experiment. Change either ingredient and the same English words can yield a different number.

The trouble grows worse the moment the words “at random” enter the picture. Consider this beautiful old problem: pick a chord of a circle “at random”; what is the probability that the chord is longer than the side of the equilateral triangle inscribed in that circle? Three perfectly

reasonable interpretations of “random chord” give three different answers.

- Pick the two endpoints of the chord uniformly and independently on the circle. The probability turns out to be $1/3$.
- Pick a radius of the circle, and then pick a point on that radius uniformly; let the chord be the one perpendicular to the radius at that point. The probability is $1/2$.
- Pick the midpoint of the chord uniformly from the disc. The probability is $1/4$.

(Working out the three numbers is a pleasant exercise once we have the right vocabulary — we will not need them today.) None of these three procedures is wrong: they are simply *three different experiments*, all wearing the same English costume. Throwing a needle, spinning a spinner against a diameter, and tossing a dart at the disc are all natural physical realisations of “a random chord”, but they realise it differently. The question “what is the probability that a random chord is longer than the side of the inscribed triangle?” has no answer until the questioner says *which* random chord.

Even classroom-scale phrases hide the same trap. “Pick a student at random from a class of thirty” is ambiguous until we specify the procedure: drawing a roll number from a bowl uniformly, picking the first person to walk in tomorrow morning, or weighting students by some criterion we did not bother to write down. Two procedures that we casually describe with the same English sentence will, in general, place different probabilities on the same event.

The escape from this fog is not cleverness; it is discipline. Before we calculate anything, we will commit, in writing, to two ingredients: a set Ω listing all conceivable outcomes of the experiment, and a rule $\mathbb{P}$ that assigns to each subset of Ω a number in $[0, 1]$ representing the likelihood that the outcome lies in that subset. Once these are fixed, every probability becomes a well-defined number, and disagreements like the one between our two students reduce to disagreements about which Ω and which $\mathbb{P}$ correctly model the experiment at hand — a far more honest place to argue. The rest of Day 1 makes this discipline precise, beginning with the language of sample spaces and events.

1.1 SAMPLE SPACES AND EVENTS

1.2 The setup. With the discipline of the previous point in mind, we now collect the formal vocabulary we will use throughout the workshop: experiment, outcome, sample space, and event.

Definition. A **random experiment** is any procedure whose outcome is not predetermined. The set of all conceivable outcomes is called the **sample space** and is usually denoted Ω (capital Omega); its elements are called **outcomes**. Any subset $A \subseteq \Omega$ is called an **event**, and we say that the event A *occurs* when the actual outcome of the experiment lies in A .

We always insist that Ω is described *before* any computation is attempted. A handful of small examples will fix the vocabulary; in each, we describe the experiment in words, write down Ω explicitly, and name a few events as subsets of Ω .

- (i) *Roll a fair six-sided die once.* The sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

The event “the roll is even” is the subset $A = \{2, 4, 6\} \subseteq \Omega$, and the event “the roll is at least 5” is $B = \{5, 6\}$. The event “the roll is a prime” is $\{2, 3, 5\}$.

- (ii) *Toss a coin three times in a row, recording the result of each toss in order.* Then

$$\Omega = \{H, T\}^3 = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\},$$

an 8-element set. The event “at least two heads” is $\{HHH, HHT, HTH, THH\}$; the event “the first toss is a tail” is $\{THH, THT, TTH, TTT\}$; the event “all three tosses are the same” is $\{HHH, TTT\}$.

- (iii) *Draw a single card from a standard 52-card deck.* The sample space Ω is the set of all 52 cards, so $|\Omega| = 52$. The event “the card is a heart” is the 13-card subset of hearts; “the card is a face card” is the 12-card subset consisting of the four jacks, four queens, and four kings; “the card is a red ace” is the 2-card subset $\{A\heartsuit, A\diamondsuit\}$.
- (iv) *Roll two distinguishable dice — say, one red and one blue — and record the ordered pair (a, b) of results.* Then

$$\Omega = \{1, \dots, 6\}^2,$$

a 36-element set. The event “the sum is 7” is

$$\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

“Both dice show the same number” is

$$\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

“The red die is strictly larger” is the 15-element set $\{(a, b) : a > b\}$.

- (v) *A family has two children; record their genders in birth order.* Then

$$\Omega = \{B, G\}^2 = \{BB, BG, GB, GG\}$$

The event “both children are girls” is $\{GG\}$; “at least one child is a boy” is $\{BB, BG, GB\}$; “the two children have different genders” is $\{BG, GB\}$.

1.3 Combining events. Because events are subsets of Ω , we can combine them with the usual set operations. To make each definition concrete, fix the running example of a single roll of a fair die, with $\Omega = \{1, 2, 3, 4, 5, 6\}$, and the two events $A = \{2, 4, 6\}$ (“the roll is even”) and $B = \{5, 6\}$ (“the roll is at least 5”).

- *Union.* $A \cup B$ — “ A or B occurs” (or both): the outcome lies in A , in B , or in both. In the running example,

$$A \cup B = \{2, 4, 5, 6\},$$

the rolls that are even or at least 5 (with 6 qualifying on both counts).

- *Intersection.* $A \cap B$ — “ A and B both occur”: the outcome lies in both A and B simultaneously. In the running example,

$$A \cap B = \{6\},$$

since 6 is the only roll that is both even and at least 5.

- *Complement.* $A^c = \Omega \setminus A$ — “ A does not occur”: the outcome lies outside A . In the running example,

$$A^c = \{1, 3, 5\},$$

the odd rolls.

Two events A and B are **mutually exclusive** (or **disjoint**) when they cannot occur together, i.e. when $A \cap B = \emptyset$. The events A and B above are *not* mutually exclusive, since their intersection contains 6; for a pair that is, let $C = \{1, 3\}$ (“the roll is odd and less than 5”) and observe that $A \cap C = \emptyset$, because no roll is both even and odd.

Exercise 1.1.1. A fair coin is tossed twice, the results recorded in order. Write down the sample space Ω , and write the event “at least one toss is a head” as a subset of Ω .

Exercise 1.1.2. For a single roll of a fair die, let $A = \{2, 4, 6\}$ and $B = \{5, 6\}$. Write down each of $A \cup B$, $A \cap B$, and A^c , and decide whether A and B are mutually exclusive.

Exercise 1.1.3. A family has three children, whose genders are recorded in birth order. Write down the sample space Ω , and write the event “exactly two of the children are girls” as a subset of Ω .

1.2 PROBABILITY MEASURES

1.4 The probability measure. A probability measure on a finite sample space Ω is, at its heart, a rule that assigns a weight to each individual *outcome* of the experiment, recording how likely that outcome is. The probability of an *event* — a subset of Ω — is then nothing more than the total weight sitting inside it. We make this precise in two steps: first the function on outcomes, then the induced function on events.

Definition. Let Ω be a finite sample space. A **probability measure** on Ω is a function

$$\mathbb{P} : \Omega \rightarrow [0, 1]$$

whose values on the outcomes sum to 1:

$$\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1.$$

The number $\mathbb{P}(\omega) \in [0, 1]$ is called the **probability mass** (or simply the **probability**) of the outcome ω .

With masses on outcomes in hand, the probability of any event $A \subseteq \Omega$ is, by definition, the sum of the masses of its outcomes:

$$\mathbb{P}(A) := \sum_{\omega \in A} \mathbb{P}(\omega).$$

We follow the standard abuse of notation and write $\mathbb{P}$ both for the function on outcomes (with input $\omega \in \Omega$) and for the induced function on events (with input $A \subseteq \Omega$); context always makes clear which is meant. Note that the original mass on a single outcome ω is recovered as the probability of the singleton event $\{\omega\}$. Three consequences of the definition come up so often that we name them explicitly.

(P1) *The total probability is 1.* Directly from the defining sum,

$$\mathbb{P}(\Omega) = \sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1.$$

In words: something always happens.

(P2) *Probabilities are non-negative.* For every event $A \subseteq \Omega$,

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega) \geq 0,$$

because each term in the sum is non-negative.

(P3) *Disjoint events add.* If A and B are disjoint events ($A \cap B = \emptyset$), then every outcome in $A \cup B$ belongs to exactly one of A or B , so splitting the sum gives

$$\mathbb{P}(A \cup B) = \sum_{\omega \in A \cup B} \mathbb{P}(\omega) = \sum_{\omega \in A} \mathbb{P}(\omega) + \sum_{\omega \in B} \mathbb{P}(\omega) = \mathbb{P}(A) + \mathbb{P}(B).$$

The same argument, applied repeatedly, gives the general form: for pairwise disjoint events $A_1, A_2, \dots, A_n$,

$$\mathbb{P}(A_1 \cup A_2 \cup \dots \cup A_n) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots + \mathbb{P}(A_n).$$

Almost every calculation in the workshop rests on (P1) and (P3); (P2) is so transparent from the definition that we record it mostly for completeness. We will refer back to these three properties by their labels throughout.

1.5 What does the probability of an event mean? The definition we have just given tells us how to *manipulate* probabilities — they are non-negative numbers, they sum to 1, they add over disjoint events — but it is silent on what the number $\mathbb{P}(A)$ is supposed to *represent* in the world. A reader who has never seen a probability measure before is well within their rights to ask: in what sense is the number $1/6$ the “right” probability of rolling a 3 on a fair die? Why this number and not some other?

The most popular answer is the **frequentist interpretation**.¹ Imagine repeating the same experiment under identical conditions a large number of times — N tosses of a coin, N rolls of a die, N random draws of a card — and recording, for each event A , the fraction

$$\hat{p}_N(A) = \frac{\#\{\text{trials in which } A \text{ occurred}\}}{N}.$$

The frequentist interpretation says: *the number $\mathbb{P}(A)$ is the long-run value of $\hat{p}_N(A)$ as N grows large.* In other words, $\mathbb{P}(A)$ is what the empirical fraction $\hat{p}_N(A)$ converges to as we run the experiment more and more times. Toss a fair coin 1000 times and you will see roughly 500 heads; toss it a million times and you will see roughly 500 000. The fraction of heads stabilises near $1/2$, and we declare $\mathbb{P}(\text{heads}) = 1/2$.

Reading the three properties (P1), (P2), (P3) through this lens, each of them becomes inevitable rather than imposed.

- (i) *Total probability is 1.* Every trial produces exactly one outcome, so the trial counts of any partition of Ω into singletons sum to N . Dividing by N , the empirical fractions across Ω sum to 1; passing to the long-run limit gives $\mathbb{P}(\Omega) = 1$, which is (P1).
- (ii) *Probabilities are non-negative.* An empirical fraction $\hat{p}_N(A)$ is a count divided by N , hence non-negative, and so its limit is non-negative too. This is (P2).
- (iii) *Disjoint events add.* If A and B are disjoint, then no single trial is counted in both: the trials in which $A \cup B$ occurred are precisely the trials in A plus those in B . Dividing by N , $\hat{p}_N(A \cup B) = \hat{p}_N(A) + \hat{p}_N(B)$ exactly. Passing to the limit gives (P3).

So the axioms are not arbitrary; they are exactly the rules that empirical fractions automatically satisfy, lifted to the long-run limit.

A word of caution. The interpretation describes *what* $\mathbb{P}(A)$ *means*; it is not a definition we can use to *compute* $\mathbb{P}(A)$, because we never actually run an experiment infinitely many times. In practice, one models the experiment by writing down a sample space and a probability measure

¹It is not the only answer. A second tradition, the *subjective* or *Bayesian* school, reads $\mathbb{P}(A)$ as a numerical measure of one’s personal degree of belief that A will occur. The two interpretations rarely lead to different calculations, and for the workshop we will be happy with the frequentist picture.

that one believes describes it correctly (often by symmetry, as for the fair coin or the fair die), and then computes with that model. The frequentist picture is the bridge between the model and reality — it explains why a number on paper deserves the name “probability”. The rest of the workshop will live almost entirely on the modelling side of the bridge; we will write down $\mathbb{P}$, manipulate it, and prove theorems about it, leaning on the frequentist intuition only when we need to remember what the symbols are about.

One final note. Even within the frequentist picture, the claim that $\hat{p}_N(A) \rightarrow \mathbb{P}(A)$ as $N \rightarrow \infty$ is itself a theorem, not a definition — this is the content of the *law of large numbers*, which we will not prove in the workshop. We will, however, see a quantitative cousin of it on Day 5: Chebyshev’s inequality, which controls how close $\hat{p}_N(A)$ is to $\mathbb{P}(A)$ for any specific finite N .

1.6 Examples of probability measures. The definition becomes much clearer in action. In each example below we exhibit a sample space Ω and the mass $\mathbb{P}(\omega)$ assigned to each outcome, check that the masses sum to 1, and — where it adds something — use the identity $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega)$ to compute the probability of a representative event.

- (i) *The fair coin.* Take $\Omega = \{\mathbf{H}, \mathbf{T}\}$ with $\mathbb{P}(\mathbf{H}) = \mathbb{P}(\mathbf{T}) = 1/2$. The masses sum to $1/2 + 1/2 = 1$, so this is a valid probability measure. The event “the coin shows heads” is the singleton $\{\mathbf{H}\}$, with $\mathbb{P}(\{\mathbf{H}\}) = 1/2$.
- (ii) *A biased coin.* On the same sample space $\Omega = \{\mathbf{H}, \mathbf{T}\}$ we could equally well take $\mathbb{P}(\mathbf{H}) = 2/3$ and $\mathbb{P}(\mathbf{T}) = 1/3$; the masses still sum to 1, so this too is a probability measure on Ω . The lesson to take away is that $\mathbb{P}$ is its own ingredient, separate from Ω : a single sample space carries infinitely many distinct probability measures, one for each physical experiment we might have in mind.
- (iii) *A fair die.* On $\Omega = \{1, 2, 3, 4, 5, 6\}$, set $\mathbb{P}(k) = 1/6$ for every $k \in \Omega$. The masses sum to $6 \cdot (1/6) = 1$. For the event $A = \{2, 4, 6\}$ (“the roll is even”),

$$\mathbb{P}(A) = \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}.$$

- (iv) *A loaded die.* Take the same sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$, but now suppose the die has been weighted so that the face 6 comes up with probability $1/2$, and each of the other five faces with probability $1/10$. Check first that the masses sum correctly:

$$\mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) + \mathbb{P}(4) + \mathbb{P}(5) + \mathbb{P}(6) = 5 \cdot \frac{1}{10} + \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1.$$

The same event $A = \{2, 4, 6\}$ (“the roll is even”) now has probability

$$\mathbb{P}(A) = \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) = \frac{1}{10} + \frac{1}{10} + \frac{1}{2} = \frac{7}{10}.$$

This is exactly the same event, on exactly the same sample space, as in the previous example — but the probability has shifted from $1/2$ to $7/10$, because the underlying mass function changed. “The probability that the die shows an even number” is not a property of the die alone; it is a property of the pair (die, weighting).

- (v) *A degenerate measure.* On $\Omega = \{a, b, c\}$, the assignment $\mathbb{P}(a) = 1$, $\mathbb{P}(b) = 0$, $\mathbb{P}(c) = 0$ has masses summing to 1 and so is a perfectly legal probability measure — though a boring one, since it describes an experiment whose outcome is a with certainty. The event $\{b, c\}$ has probability $\mathbb{P}(b) + \mathbb{P}(c) = 0$: even though b and c appear in the sample space, this particular measure rules them out, and outcomes of mass zero contribute nothing to the probability of any event that contains them.

1.7 Consequences of the rules. A few simple consequences will be used constantly.

Elementary properties. Let $A, B \subseteq \Omega$ be events. Then

- (i) $\mathbb{P}(\emptyset) = 0$;
- (ii) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
- (iii) if $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$;
- (iv) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
- (v) $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$ (**union bound**).

Proof. (i) The empty set is disjoint from itself, so by (P3), $\mathbb{P}(\emptyset) = \mathbb{P}(\emptyset \cup \emptyset) = 2\mathbb{P}(\emptyset)$, forcing $\mathbb{P}(\emptyset) = 0$. (ii) Since A and A^c are disjoint and $A \cup A^c = \Omega$, we have $\mathbb{P}(A) + \mathbb{P}(A^c) = 1$. (iii) Write $B = A \cup (B \setminus A)$ as a disjoint union; then $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$. (iv) Decompose $A \cup B$ as the disjoint union $A \cup (B \setminus A)$, and decompose B as the disjoint union $(B \setminus A) \cup (A \cap B)$. Adding gives the claim. (v) Follows from (iv) since $\mathbb{P}(A \cap B) \geq 0$. ■

The union bound (v) extends to any number of events: $\mathbb{P}(A_1 \cup \dots \cup A_n) \leq \mathbb{P}(A_1) + \dots + \mathbb{P}(A_n)$. Despite (or because of) its simplicity, the union bound will be one of the workhorses of the workshop.

Exercise 1.2.1. A die is loaded so that the face 6 comes up with probability $1/2$, and each of the other five faces with probability $1/10$. Find the probability that the roll is odd, and the probability that the roll is at least 5.

Exercise 1.2.2. Let A and B be events with $\mathbb{P}(A) = 0.6$, $\mathbb{P}(B) = 0.5$, and $\mathbb{P}(A \cap B) = 0.2$. Find $\mathbb{P}(A \cup B)$ and $\mathbb{P}(A^c)$.

Exercise 1.2.3. Using the two-event union bound $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$, show that for any three events A , B , and C ,

$$\mathbb{P}(A \cup B \cup C) \leq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C).$$

1.3 THE UNIFORM PROBABILITY MEASURE

1.8 Equally likely outcomes. When the sample space Ω is finite and we model each outcome as “equally likely”, the probability measure is uniquely determined: each singleton $\{\omega\}$ must receive probability $1/|\Omega|$. From (P3), the probability of any event is then

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{\#\{\text{favourable outcomes}\}}{\#\{\text{total outcomes}\}}.$$

This is the **uniform probability measure**, sometimes also called the *equiprobable model*. Almost every problem on the first day will use this model.

Example 1.9. Roll a fair six-sided die twice; record the ordered pair (a, b) of results. Then $\Omega = \{1, \dots, 6\}^2$, so $|\Omega| = 36$. The event $A = \{a + b = 7\}$ contains the six outcomes $(1, 6), (2, 5), \dots, (6, 1)$, so under the uniform measure $\mathbb{P}(A) = 6/36 = 1/6$.

Example 1.10. A box contains 5 red and 3 blue balls; two balls are drawn without replacement. What is the probability that both are red?

Take Ω to be the set of unordered pairs of balls; $|\Omega| = \binom{8}{2} = 28$. The number of unordered pairs of red balls is $\binom{5}{2} = 10$. So the probability is $10/28 = 5/14$.

There is a second, equally valid sample space: ordered pairs of distinct balls. Then $|\Omega| = 8 \cdot 7 = 56$ and the favourable count is $5 \cdot 4 = 20$, giving the same answer $20/56 = 5/14$. This is reassuring: the answer should not depend on whether we record order, provided we are consistent.

1.11 A Worked Example: The Birthday Problem. Probability under the uniform model reduces to counting, and we will lean on a small set of standard counting principles — multiplication, permutations, k -permutations, and combinations (binomial coefficients) — throughout this workshop. These are collected, with their derivations, in Appendix ??; readers who are not yet comfortable with any of the four should pause and skim that appendix before continuing. A first application of these principles, big enough to be interesting and small enough to be solved in one move, is the famous birthday problem.

A classroom contains 30 students. Assuming each student's birthday is equally likely to be any of 365 days, and birthdays are chosen independently, what is the probability that some two students share a birthday? The cleanest sample space is $\Omega = \{1, \dots, 365\}^{30}$, with the uniform measure; an outcome is an ordered list of 30 birthdays, one per student. So $|\Omega| = 365^{30}$. The complement of “some two share a birthday” is “all 30 birthdays are distinct”. Counting the ordered 30-tuples of *distinct* birthdays gives

$$365 \cdot 364 \cdot 363 \cdots 336 = \frac{365!}{335!}.$$

So

$$\mathbb{P}(\text{all distinct}) = \frac{365 \cdot 364 \cdots 336}{365^{30}} \approx 0.294,$$

and the probability of a shared birthday is about 0.706. The lesson, which will reappear: with 30 outcomes among 365 slots, “collisions” are far more likely than the size of the room alone might suggest.

Exercise 1.3.1. A fair die is rolled twice. Find the probability that the sum of the two rolls equals 5, and the probability that the sum is even.

Exercise 1.3.2. A box contains 4 red balls and 2 green balls. Two balls are drawn without replacement. Find the probability that both are red.

Exercise 1.3.3. A number is chosen uniformly at random from $\{1, 2, \dots, 10\}$. Find the probability that it is prime.

1.4 INDEPENDENCE OF EVENTS

1.12 Independence. Two events A and B are called **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

Intuitively: knowing whether A occurred does not change the probability that B occurs. The definition is what it is — a *multiplication rule for the joint probability* — and we will use it in that form, without philosophy.

More generally, events $A_1, A_2, \dots, A_n$ are **mutually independent** if for every subset of indices $i_1 < \dots < i_k$,

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \mathbb{P}(A_{i_2}) \cdots \mathbb{P}(A_{i_k}).$$

(One must check the multiplication rule for *every* subset, not only for pairs.)

Example 1.13 (Position-fixed events in a random permutation). Let π be a uniformly random permutation of $\{1, 2, \dots, n\}$, with $n \geq 2$. For each i , let

$$A_i = \{\pi(i) = i\}$$

be the event “ i is a fixed point of π .” These events look as if they ought to be independent — the position $\pi(1)$ takes seems to have nothing to do with the position $\pi(2)$ takes — and yet they are not.

Work in $\Omega = S_n$ (all $n!$ permutations, with the uniform measure). The number of permutations fixing $\pi(1) = 1$ is $(n-1)!$ (we are free to permute the remaining $n-1$ values), so

$$\mathbb{P}(A_1) = \frac{(n-1)!}{n!} = \frac{1}{n}.$$

Similarly, the number of permutations fixing both $\pi(1) = 1$ and $\pi(2) = 2$ is $(n-2)!$, and so

$$\mathbb{P}(A_1 \cap A_2) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}.$$

If A_1 and A_2 were independent, the joint probability would factor as $\mathbb{P}(A_1)\mathbb{P}(A_2) = 1/n^2$. Instead,

$$\frac{1}{n(n-1)} > \frac{1}{n^2}.$$

Knowing $\pi(1) = 1$ makes $\pi(2) = 2$ *more* likely, because once the value 1 has been used at position 1, the value 2 has one fewer competitor for position 2. The bias is small for large n but it is real, and it is exactly the kind of dependence that informal reasoning tends to miss.

1.14 A first useful identity. A typical use of mutual independence is when we want to argue that “none of a long list of bad things happens”. Applying the multiplication rule to the complementary events — i.e., to the events “each individual bad thing does *not* happen” — turns this into a product, which is often vastly easier to estimate than the original probability.

Probability of avoiding many events. If $A_1, \dots, A_n$ are mutually independent, then

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i^c\right) = \prod_{i=1}^n \mathbb{P}(A_i^c) = \prod_{i=1}^n (1 - \mathbb{P}(A_i)).$$

Proof. It is a (standard but non-trivial) fact that if $A_1, \dots, A_n$ are mutually independent then so are $A_1^c, \dots, A_n^c$. The identity follows by applying the multiplication rule. ■

A useful consequence we will return to repeatedly: for any reals x_i ,

$$\prod_{i=1}^n (1 - x_i) \leq \exp\left(-\sum_{i=1}^n x_i\right),$$

since $1 - x \leq e^{-x}$ for all real x . So even when the union bound is too crude, an independence argument can produce sharp exponential decay.

Exercise 1.4.1. Events A and B are independent, with $\mathbb{P}(A) = 1/2$ and $\mathbb{P}(B) = 1/3$. Find $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A \cup B)$.

Exercise 1.4.2. A fair coin is tossed three times, so that $\Omega = \{\text{H}, \text{T}\}^3$ carries the uniform measure. Let A be the event “the first toss is H” and B the event “the second toss is T”. By counting outcomes, determine whether A and B are independent.

Exercise 1.4.3. Let $A_1, \dots, A_n$ be mutually independent events, each of probability p . Write the probability that none of them occurs as a product, and use the inequality $1 - x \leq e^{-x}$ to give an upper bound for it.

1.5 PUTTING IT TOGETHER

1.15 The shape of every probability problem. Every problem we solve in this workshop — whether elementary or advanced — has the same shape:

- (1) **Choose a sample space** Ω that captures the experiment.
- (2) **Choose a probability measure** $\mathbb{P}$ on Ω (most often, the uniform measure on a cleverly chosen Ω).
- (3) **Identify the event** A whose probability you want.
- (4) **Compute** $\mathbb{P}(A)$, often by counting or by combining simpler probabilities using the rules above.

The *art* lies in steps (1) and (2): the right Ω can turn an unwieldy question into a one-line calculation. Step (4) is then mostly a careful count.

PROBLEM SET 1

The following problems are to be attempted before tomorrow’s session. They consolidate today’s definitions (sample spaces, events, the uniform measure, counting, and independence) and prepare the ground for the language of expectation, which we will introduce on Day 2.

Problem 1.5.1. A fair coin is tossed four times in a row.

- (a) Write down a sample space Ω for this experiment, and state the probability measure $\mathbb{P}$ on Ω .
- (b) Compute $\mathbb{P}(\text{exactly two heads})$.
- (c) Compute $\mathbb{P}(\text{the first toss is a head})$.
- (d) Compute $\mathbb{P}(\text{the first and last tosses agree})$.

Problem 1.5.2. Five distinct cards are drawn from a standard 52-card deck.

- (a) How many distinct 5-card hands are there?
- (b) What is the probability that all five cards are spades?
- (c) What is the probability that the hand contains exactly one ace?

Problem 1.5.3. Show, directly from the three rules (P1)–(P3), that for any events $A_1, A_2, \dots, A_n$,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

This is the *union bound*; we will use it many times.

Problem 1.5.4. Twenty people are seated at random around a circular table. What is the probability that two particular people, Alice and Bob, sit next to each other?

Problem 1.5.5. Six six-sided dice are rolled simultaneously. Compute the probability that

- (a) all six show the same value;
- (b) all six values are different;
- (c) at least two dice show the same value.

Problem 1.5.6. Let $\Omega = \{1, 2, 3, 4\}$ with the uniform probability measure. Define the events

$$A = \{1, 2\}, \quad B = \{1, 3\}, \quad C = \{1, 4\}.$$

- (a) Show that A, B, C are *pairwise* independent.
- (b) Show that A, B, C are *not* mutually independent.

(This shows that pairwise independence is strictly weaker than mutual independence.)

Problem 1.5.7. A coin with probability p of heads is tossed n times independently. Express the probability that the resulting string contains at least one head. Now use the inequality $1 - x \leq e^{-x}$ to show that for any $p \in (0, 1)$, this probability exceeds $1 - e^{-pn}$.

This is our first “meet a threshold” calculation: for a small p , we need only $n \approx 1/p$ tosses before a head becomes very likely.

Problem 1.5.8. A standard 52-card deck is shuffled uniformly at random. What is the probability that the Ace of Spades appears earlier in the shuffled order than the Ace of Hearts?

Problem 1.5.9. Three friends — Alice, Bob, and Carol — leave their wallets at the door of a party. As they leave, an inattentive attendant returns the three wallets to the three friends in a uniformly random order.

- (a) Compute the probability that every friend gets their own wallet.
- (b) Compute the probability that *nobody* gets their own wallet.
- (c) Compute the probability that exactly one friend gets their own wallet.
- (d) Could exactly two friends get their own wallets? Explain.

Problem 1.5.10. The $2n$ players of a tennis tournament are paired up uniformly at random into n matches (so the sample space is the set of all ways to partition $\{1, 2, \dots, 2n\}$ into n unordered pairs, with the uniform measure on it). What is the probability that players 1 and 2 are paired against each other?

Problem 1.5.11. A fair coin is tossed five times, producing a string in $\{\text{H}, \text{T}\}^5$.

- (a) Compute the probability that the string contains no two consecutive Hs (i.e., no HH substring).
- (b) Let a_n denote the number of length- n strings in $\{\text{H}, \text{T}\}^n$ with no HH substring. By splitting on the last character, show that

$$a_n = a_{n-1} + a_{n-2} \quad \text{for } n \geq 3.$$

(Do you recognise the sequence?)